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Ministry of Labor and Skills

Animal Health

Level-I

Based on March 2022, Version 4 Occupational standard

Module Title: - Identifying and Handling Basic Veterinary Tools and Equipment

LG Code: AGR ANH1 M01 LO (1-3) LG (1-3)

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Introduction to the Module

This module covers the knowledge, skills and attitude required to identify and operate with basic tools and equipment. The module also requires the application of skills and knowledge to a specified range of tasks to clean and store basic veterinary tools and equipment. In addition, the competence requires an awareness of workplace safety and positive environmental practices associated with equipment operation

LG #1	LO #1- Identify, Prepare and use basic tools and equipment

Instruction sheet 1
This learning guide is developed to provide you the necessary information regarding the following content coverage and topics: • Basic tools, equipment and machinery in animal health care and service • pre-operational checks of basic veterinary tools and equipment • Performing simulation trial to operate basic tools and equipment • Identify, segregate, and report unsafe or non-functional tools and equipment • Understanding of Occupational health and safety (OHS) • Work place hazards and Safe Work Procedures This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to: • Identify, select use maintain, and store. Suitable PPE, Tools, and equipment based on instructions and OHS requirements • Understand the use of basic veterinary tools and equipment • Carry out the Routine pre-operational checks of tools and equipment based on the manufacturer's specifications. • Carry out a simulation trial is on how to operate basic tools and equipment. • Identify, segregate, and report Unsafe or non-functional tools and equipment for repair or replacement. • Identify and report OHS hazards in the workplace to the supervisor.
Learning Instructions:

1. Read the specific objectives of this Learning Guide.
2. Follow the instructions described below.
3. Read the information written in the information Sheets
4. Accomplish the Self-checks
5. Perform Operation Sheets
6. Do the "LAP test"

Information Sheet 1:

1.1. Basic tools, equipment and machinery in animal health care and service

In the veterinary world tools and equipments are simplify and assist the veterinarian to perform different tasks to control and prevent animal diseases and maximize production

Definition of terms in veterinary tools and equipments

- **Materials: -** are things needed for an activity, cleaning materials or consumable materials. E.g. soap, salon, mop etc.
- **Tools: -** is any physical item that can be used to achieve a goal, especially if the item is not consumed in the process.
- **Equipments: -** is the necessary item for a particular purpose. E.g. centrifuge, autoclave etc.
- **Instrument: -** device that communicates**,** denotes, detects, indicates
- **Measures**, observes records or signals a quantity or phenomenon, or controls or manipulates another device.
- **Machine**: - is a tool containing one or more parts that use energy to perform an intended action. It is usually powered by mechanical, chemical, thermal or electrical means, and is often motorized.
- **Device: -** Machine designed for a purpose. In a general context computer is device.
- **Utensil: -** A simple and useful device that is used for doing tasks in a person's home and especially in kitchen.
- **Object: -** something perceptible by one or more of the senses, especially by vision or touch, a material thing.
- **Apparatus:** A group or combination of instruments, machinery, tools, materials, etc., having a particular function or intended for a specific use.

1.1.1. Personal protective equipments used in animal health.

Personal protective equipment, commonly referred to as "PPE", is equipment worn to minimize exposure to hazards that cause serious workplace injuries and illnesses. These injuries and illnesses may result from contact with chemical, radiological, physical, electrical, mechanical, or other workplace hazards. Personal protective equipment may include items such as gloves, safety glasses and shoes, earplugs or muffs, hard hats, respirators, or coveralls, vests and full body suits.

What can be done to ensure proper use of personal protective equipment?

All personal protective equipment should be safely designed and constructed, and should be maintained in a clean and reliable fashion. It should fit comfortably, encouraging worker use. If the personal protective equipment does not fit properly, it can make the difference between being safely covered or dangerously exposed. When engineering, work practice, and administrative controls are not feasible or do not provide sufficient protection, employers must provide personal protective equipment to their workers and ensure its proper use. Employers are also required to train each worker required to use personal protective equipment to know:

- When it is necessary
- What kind is necessary
- How to properly put it on, adjust, wear and take it off
- The limitations of the equipment
- Proper care, maintenance, useful life, and disposal of the equipment

If PPE is to be used, a PPE program should be implemented. This program should address the hazards present; the selection, maintenance, and use of PPE; the training of employees; and monitoring of the program to ensure its ongoing effectiveness.

Types of PPE

- **Gloves**:
    - ***Purpose:*** Gloves are used to prevent hand contamination and the spread of infection between animals and off the property.
    - **Type of gloves includes:**
        - ✓ Reusable heavy duty gloves
        - ✓ Disposable latex, nitrite or vinyl gloves
        - ✓ Cut resistant gloves.

- **How to use gloves?**

Practical considerations when using gloves include:

- ✓ If using duct tape to seal gloves to overalls, make sure this does not tear the gloves or overalls during removal as this may cause contamination.
- ✓ Consider using extended cuff gloves to protect the wrist.
- ✓ Change gloves if they become torn or damaged during use.
- ✓ Always perform hand hygiene immediately after removing gloves.
- ✓ If using latex gloves, select a low protein, powder-free make to minimize the risk of latex allergy.

Figer 1.1 Surgical, Leather gloves and Arm length glove

Source (https://www.indiamart.com/proddetail/arm-length-veterinary-gloves-17058192355.html) (acesse s on 26/08/2022)

- **Eye and face protection**
  - **Purpose**: Eye and face protection is used to prevent contamination of the face and mucous membranes
  - **Type of eye and face protection includes**:
    - ✓ Surgical/medical mask (respirator).
    - ✓ Safety glasses
    - ✓
    - ✓ safety goggles
    - ✓ Face shield
- **Protective clothing**
  - ***Purpose of protective clothing:*** is used to prevent contamination of the skin and personal clothing and the spread of infection between animals and off the property
  - ***Type of protective clothing includes*:**
    - ✓ reusable cloth overalls (coveralls)

- ✓ disposable gowns
- ✓ disposable overalls

Figure. 1.2. Cover all and Gown

Source (https://www.shaigoocare.com/isolation-gowns-coveralls/)(acesses on 26/08/2022)

- **Respiratory protective equipment (RPE)**
    - ***Purpose of Particulate RPE***: is used to prevent inhaling infectious aerosols and other particles and contamination of the nose and mouth
    - ***Type of RPE includes*:**
        - ✓ Disposable P2 respirator
        - ✓ Reusable half-face piece respirator with a particulate filter
        - ✓ Reusable full face piece respirator with a with a particulate filter
        - ✓ Powered air purifying respirator (PAPR) with a particulate filter.

Figer 1.3. Nose and face mask

Source (https://www.worksafe.qld.gov.au/safety-and-prevention/creating-safe-work/managing-risks/personal-protective-equipment-ppe/respiratory-protective-equipment-rpe) (acesse s on 26/08/2022)

(https://gzyyppe.en.made-in-china.com/product/NOcnmIQxRaWJ/China-101304-Safety-Face-Shield-Face-Shield-Visor-Safety-Face-Shield-Protective-Face-Shield.html) (acesse s on 26/08/2022)

- **Protective footwear**
  - **Purpose of protective foot wear**: is used to prevent contamination of everyday footwear and the spread of infection between animals and off the property
  - **Type of protective footwear includes**: shoe/boot covers and rubber boots.

Personal Protective Equipme...

Figer 1.4. Foot wear

Source (https://www.workplacepub.com/ppe/foot/best-foot-forward-safety-footwear/) (accesses on 26/08/2022)

- **Hat: Used to protect from sun and cold condition**

Sun Protection Hat: UV50+ U...
e4hats.com

The Best Sun Hats for Women According ...

Figure. 1.5. Hats

https://www.worksafe.qld.gov.au/safety-and-prevention/hazards/hazardous-exposures/biological-hazards/diseases-from-animals/hendra-virus/hendra-virus-information-for veterinarians/selecting-and-using-ppe-in-veterinary-practice (accesses on 26/08/2022)

1.1.2. Tools and equipments used for diagnosis of animal disease.

A. Tools used for clinical diagnosis

- **Stethoscope**
  - ✓ A stethoscope is a medical instrument used to listen to sounds produced in the body, especially those that emanate from the heart and lungs.
  - ✓ Most modern stethoscopes are binaural; that is, the instrument is intended for use with both ears.
  - ✓ Stethoscopes comprise two flexible rubber tubes running from a valve to the earpieces.
  - ✓ The valve also connects the tubes to the chest piece, which can be either a bell-shaped piece to pick up low sounds or a flat disk for higher frequencies.
  - ✓ The stethoscope is used mainly for the detection of heart murmurs, irregular heart rhythms, or abnormal heart sounds. It is also used to listen to the sound of air moving through the lungs in order to detect abnormalities in the air tubes and sacs found in the lung walls.

Parts of stethoscope

a. Earpiece: The earpiece is softer knob-like objects that are put into the ear canals so that sound can be heard.

b. Ear Tubes: Ear rubes are two metal pieces that hold the earpieces and attach to the tubing.

c. Tubing: Tubing carries the sounds from the chest piece up to the earpieces.

d. Piece: The chest piece, also known as the head, can be single-sided or double-sided. It is placed on the patient and captures the sounds that are transmitted up and through the earpieces.

e. Chill Ring: The chill ring provides protection from a cold chest piece being placed directly on a patient's skin

Figure 1.6:- picture of stethoscope

Source (https://slidingmotion.com/parts-of-stethoscope-functions/(assessed date on 29/08/2022)

- **Thermometer**

  How to use thermometer?

  Sourec ((https://www.youtube.com/watch?v=EHAmRPqi3Ks(accesses on 26/08/2022)

  ✓ Serve as a key tool in veterinary technician's to measure body temperature.

  ✓ Most vet techs use digital, as opposed to analog, thermometers, to allow for a quick temperature reading.

Figure 1.7:-Analogue and Digital Thermometers

https://www.amazon.com/Thermometer-Electronic-Underarm-Readings-Children/dp/B0007RTEAI(acessed date 29/08/2022)

- **Heart girth:** Used to measure the body weight of animals

Figer. 1.8 Heart girth meter and site of measurement

Source (https://cals.arizona.edu/backyards/sites/cals.arizona.edu.backyards/files/p11-12.pdf) accessed date 29/08/2022)

- **Ultrasound: -**

It is device that uses high frequency sound waves to capture live images from the inside of body. It can provide view of the bladder, brain (in infants), eye, gall bladder, kidney, liver, ovaries, pancreases, spleen thyroid, testicles, uterus, and blood vessels.

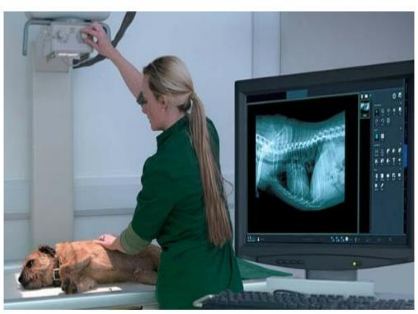
Veterinary Ultrasound | Dog Ultrasound ...
lucernevet.com

Figure 1.9. Ultrasound

Source (https://www.lucernevet.com/services/veterinary-ultrasound(29/08/2022)

- **X-Ray machine**

7- Digital X-Ray machines



Digital radiology or radiography (DR) can help the vet obtain a clear picture of the muscles, bones, and internal organs without using film and darkroom processing.

animal xray - Google Search for...
ro.pinterest.com

Figure 1.10. Digital X-Ray machine

Source (https://ro.pinterest.com/pin/191262315401572887/(assessed date 29/08/2022)

B. Veterinary. Laboratory Tools and Equipments

- **Autoclave/steam under pressure**
  - ✓ It is an electrically operated instrument used to steralize equipments and supplies by subjecting them to high pressure saturated steam at 121^{o}c for around 15-20 minute.
  - ✓ It can be used to steralize materials which can not with stand the higher temprature of hot air oven (media, rubber materials……).

Parts of autoclave

a. **Steam valve**: used to out flow of stem on autoclave¬

b. **Pressure gauge:** control the autoclave pressure¬

c. **Screw caps/lids**: used to closing of autoclave¬

d. **Perforated metallic basket**: holding of material to be sterilize in autoclave¬

e. Pressure cooker- used to boiling of water in the autoclave

Figure 1.11:- picture of autoclave

Source (https://microbenotes.com/autoclave/ accessed date 29/08/2022)

- **Centrifuge**
  - ✓ It is a rotating apparatus.

- ✓ It is used for separating objects of different densities usually from a liquid medium.
- ✓ Different types of centrifuges are used for different purposes namely hematocrite centrifuge, which can be used for centrifugation of blood contained in the capillary tube, and test tube centrifuge, which can be used to centrifuge a sample in the test tube.
- ✓ A centrifuge can be rotated at different speed to form precipitate at the base of the test tube and the solution on the top of the precipitate is termed supernatant.

Figure 1.12.; Different types of centrifuge

Source (https://www.istockphoto.com/photos/centrifuge (accesses on 26/08/2022)

- **Vacutainer tube and Capillary tube**

✓ Vacutainer tube is an instrument used to take/hold a blood sample. There are two types of vacutainer tube

- Heparinized:- has anticoagulant
- Non heparinized:- no anticoagulant

- ✓ Capillary tube is used to take blood from pierced ear vein or from blood holding vacutainer tube.

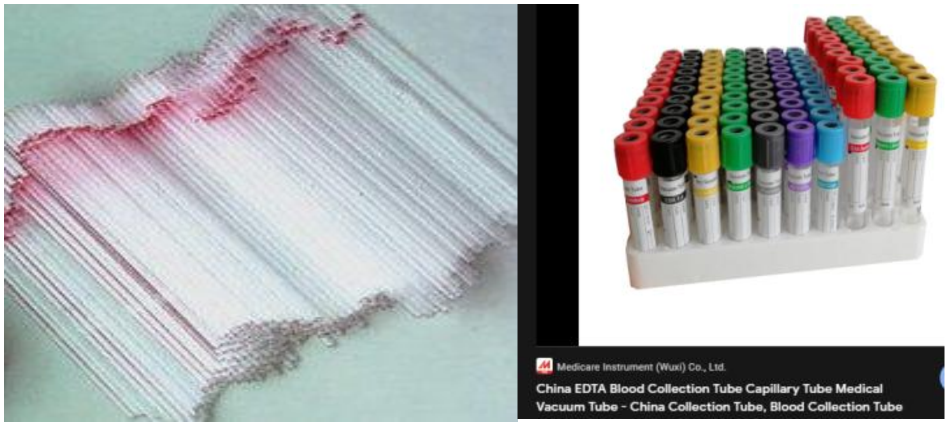


Figure 1.13:- Pictures of Vacutainer and capillary tubes

https://chinamedicare.en.made-in-china.com/product/JZlTiOKcEVYq/China-EDTA-Blood-Collection-Tube-Capillary-Tube-Medical-Vacuum-Tube.html(29/08/2022)

- **Universal bottle/plastic container**
  - ✓ Used for sample collection (faeces, ectoparasite, adult parasite……).

Figure 1.14:-Sample collection bottle

Source (https://www.123rf.com/photo_9054873_one-empty-an-sterile-specimen-collection-bottle-the-other-bottle-is-filled-with-urine-ready-for-test.html(29/08/2022)

- **Refrigerator**
  - Used for refrigeration of asubstance
  - All the container stored in refrigerators should be clearly labeled with the scientific names of the contents, the date stored and the name of the individual who stored them.
  - Flammable solutions must not be stored in arefrigerator unless it is explosion proof.

Figure 1.15:- Picture of refrigerator

Source (https://www.indiamart.com/proddetail/deep-refrigerator-14925885455.html(29/08/2022)

- **Water bath**

Water bath is a scientific instrument used for regulating the temperature of substances subjected to heat. It is also used to sterilize needles, syringes and maintain Medias as liquid. The vessel of the water bath equipment is surrounded by another vessel containing water which can be kept at a desired temperature. Water baths for laboratory are used to heat those substances, which can't be heated directly on Bunsen burner or hot plate or any other such media. However only those materials can be heated with laboratory water bath whose boiling point is less than that of water.

Figure 1 16:- picture of water bath

Source (https://www.sciencephoto.com/media/731245/view/electronic-laboratory-water-bath(29/08/2022)

- **Water distiller**
  - ✓ Water Distiller is a piece of equipment that is used to produce high-quality water through the process of Distillation.
  - ✓ Distillation is the process of removing contaminants such as heavy metals and salts that boiling water leaves behind.

Figer:1.17. Water distiller apparatus

Source ; https://tcyiqi2021.en.made-in-china.com/product/WZzTbngKJfru/China-Laboratory-Water-Distillation-Apparatus-Stainless-Steel-Multifunctional-Distilled-Water-Machine-Price.html 30/08/2022

- **Weighing Balance**
  - ✓ There are different types of weighing balances which are used in laboratories.
  - ✓ A physical balance is commonly used in the laboratories.
  - ✓ However, more accurate weighing is done by microbalances.
  - ✓ These balances are covered within a glass cover.

Figure 1.18:- picture of weighing balance

Source (https://www.kingscientific.co.uk/laboratory/Balances(29/08/2022)

- **Hot air oven**
  - ✓ Is an electrical devices used for sterilization of equipments
  - ✓ It use dry heat to sterilize equipments
  - ✓ It can be operated from 50 to 300^{o}c
  - ✓ Used to sterilize articles that can with stand high temperature and not get burnet, like glassware and powders.

Figure 1.19:- Picture of dry heat oven

Source (https://microbeonline.com/hot-air-oven-parts-types-and-uses/(29/08/2022)

- **Microscopes**
  - ✓ It is a scientific instrument with one or more lenses used to observe things not visible to the naked eye, e.g. microorganisms (bacteria, virus, fungus, etc.)
  - ✓ Is an instrument used to obtain a magnified image of minute objects?

Figure 1.20- Binocular microscope and Mononocular microscope

Sourece (https://otosection.com/types-of-microscopy-types-of-microscopes/(accesses on 26/08/2022)

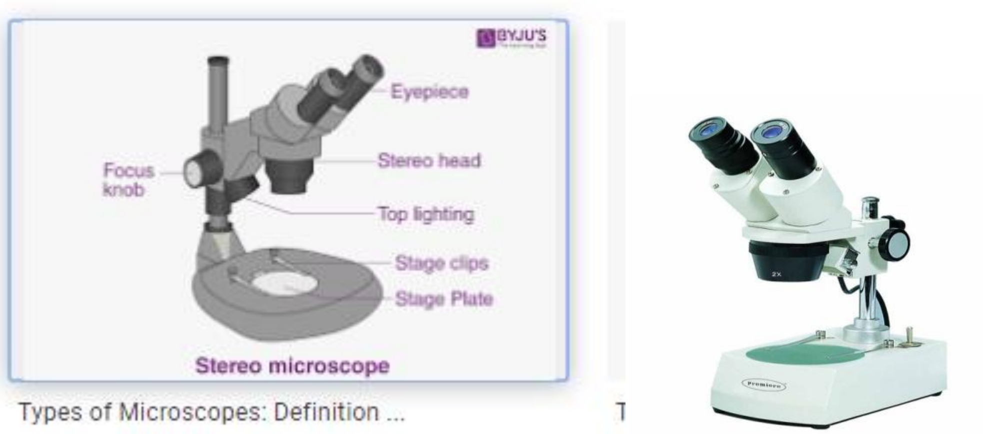


Figure 1.21:- Stero Microscope:- is an instrument used to observe Ectoparasite or the identification of EctoParasite

Source (https://byjus.com/physics/types-of-microscope/(accessed date 29/08/2022)

Parts of Microscope☹

(https://www.youtube.com/watch?v=SUo2fHZaZCU)(accesses on 26/08/2022

The Parts of Microscope Eyepiece Lens: the lens at the top that you look through. It re-magnifies the image formed by the objective lens. They are usually 10X or 15X power. Body tube: Connects the eyepiece to the objective lenses. It transmits the image from objective lens to the ocular lens. Arm: Supports the body tube and ocular lens and connects it to the base: The bottom of the microscope, used for support

a) **Eyepiece lens/ocular and objective lenses** are able to zoom in and out to enlarge and observe the specimens more clearly, with the objective lenses having a much better zoom-in ability.
b) The **eyepiece lens** is at the top where you look into to observe the specimen & able to enlarge the object 10x.
c) **Objective lenses** are usually 3 or 4 on a microscope held by a revolving nosepiece/turret, - consisting of different times of zoom in, e.g. 4x, 40x, 100x, 400x....
d) **Nosepiece/turret used** to hold and rotate objective lenses.
e) **stage clipper** used to secure/hold slide containing the specimen on the stage
f) **stage control knob** helps to move the mark on the slide into the light path
g) **Coarse focus/coarse adjustment** are used for moving the objective lenses nearer/further away from the specimen.
h) **Fine focus/ fine adjustment** are used for fine-tuning the focus (usually when it is unclear).
i) Use the coarse focus before the fine focus unless there is no need to zoom in on the specimen.
j) **The illuminator** shines a light on the specimen so that you can see it.
k) **The iris diaphragm** controls the cone of light that reaches the objectives

l) **Power on/off button** the microscope runs on power so when it is switched on, the light will also turn on

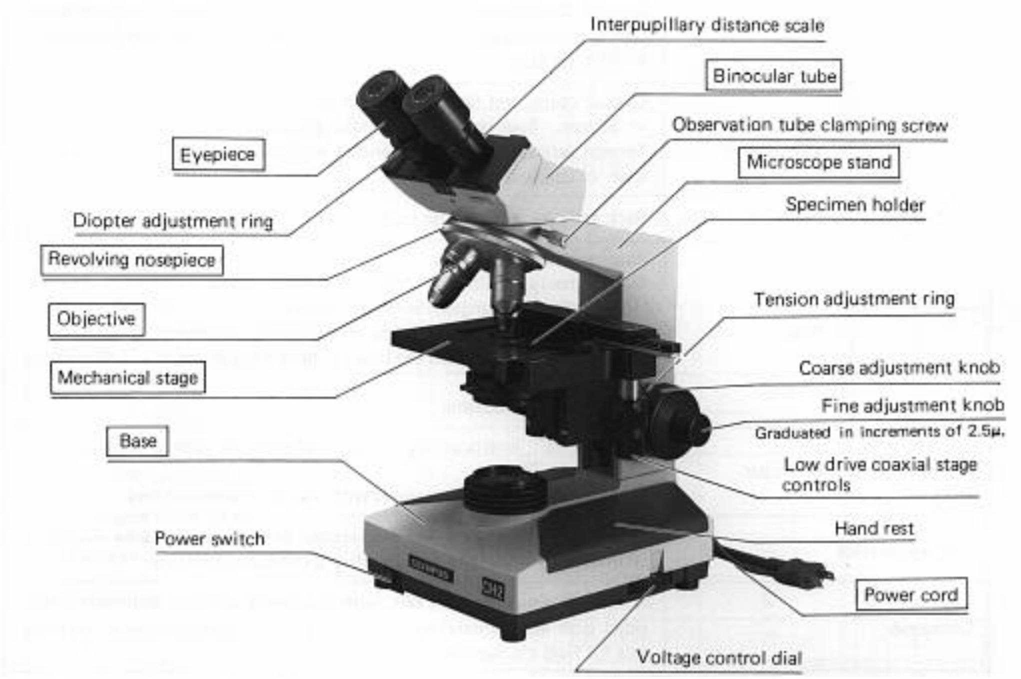


Figure 1.22. Parts of the microscope.

Source: https://www2.hawaii.edu/~johnb/micro/m140/syllabus/week/handouts/m140.2.4.html (Accessed on 25/08/2022

- **Bio safety cabinet**
  - ✓ Effective primary containment device used in laboratory working with infectious agents.
  - ✓ Prevent the escape of biological aerosols to laboratory environment.
  - ✓ It is designed for personal protection, product and environmental protection.

Figure 1.23. Bio safety cabinet

Source (https://microbenotes.com/biosafety-cabinets/(accessed date 29/08/2022)

- ***Incubator: -***
  - ✓ An incubator is a device used to grow and maintain microbiological cultures or cell cultures.
  - ✓ The incubator maintains optimal temperature, humidity and other conditions such as the carbon dioxide (CO_2) and oxygen content of the atmosphere inside.

.

Figure 1.24. Incubators

Source()(rubM54Q&ust=1661859355460000&source=images&cd=vfe&ved=0CAkQjRxqFwoTCJi16JH76_kCFQAAAAAdAAAAABAD (accessed on 29/08/2022)

- **Petri dish** is a shallow, cylindrical, round glass dish, which is used to culture different microorganism and cells. It is one of important laboratory equipments used study microorganisms like bacteria and virus in detail.

Figer 1.25. Petri dish

Source ; https://www.ubuy.vn/en/product/URSO3E-pyrex-3160-101-brand-3160-petri-dish-100-x-15-mm-pack-of-12-3455103(29/08/2022)

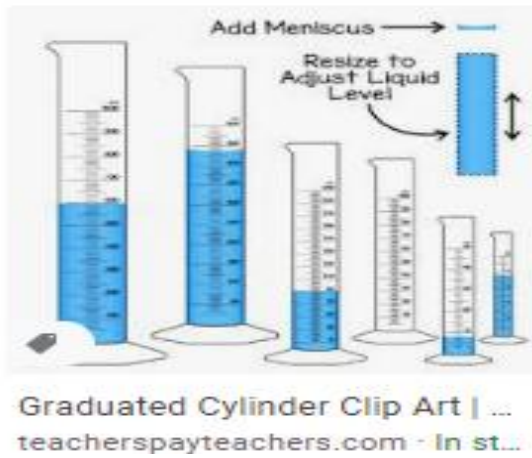
- **Beaker:** A beaker is a cylindrical container used to store, mix and heat liquids in laboratories. Most are made of glass, but other non-corrosive materials, such as metal and heat-resistant plastic, are also used.

Figer 1.26.: beaker and flask

Source ;(https://www.indigoinstruments.com/glassware/beakers/500ml-beaker-glass-griffin-55110.html accesses on 29/8/2022)

- **Graduated cylinder**

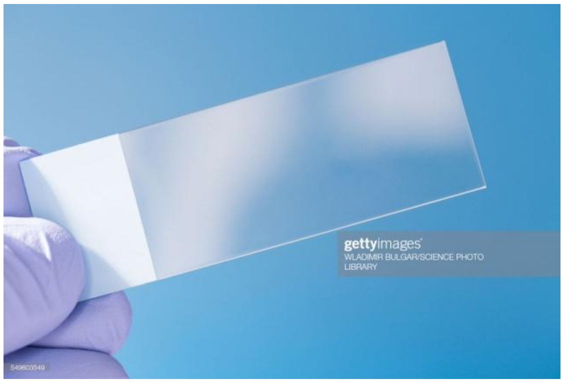
- ✓ Graduated cylinder, measuring cylinder or mixing cylinder is a common piece of laboratory equipment used to measure the volume of a liquid.
- ✓ It has a tall, narrow cylindrical shape with a scale, or graduations mark.
- ✓ Each marked line on the graduated cylinder represents the amount of liquid that has been measured.



Figer: 1.27: graduated cylinder

Source: https://www.teacherspayteachers.com/Product/Graduated-Cylinder-Clip-Art-Measuring-Volume-6410905(29/08/2022)

- **Microscope slide**
  - ✓ A **microscope slide** is a thin flat piece of glass, typically 75 by 26 mm (3 by 1 inch) and about 1 mm thick, used to hold objects for examination under a **microscope**



- **Microscope slide cover slip**
  - ✓ This smaller sheet of glass, called a **cover slip** or cover glass, is usually between 18 and 25 mm on a side. the cover glass serves for two purposes:
  - ✓ It protects the **microscope's** objective lens from contacting the specimen, and
  - ✓ It creates an even thickness (in wet mounts) for viewing.

Figure: 1.28: Microscopic slide and cover slip

Source: (https://www.sciencecompany.com/Microscope-Slide-Cover-Glass-22-x-22mm-P5930(acessess on 29/8/2022)

- **Pipette**

- ✓ A pipette (sometimes spelled pipet) is a laboratory tool commonly used in chemistry, biology and medicine to transport a measured volume of liquid, often as a media dispenser.
- ✓ Pipettes come in several designs for various purposes with differing levels of accuracy and precision, from single piece glass pipettes to more complex adjustable or electronic pipettes. Many pipette types work by creating a partial vacuum above the liquid-holding chamber and selectively releasing this vacuum to draw up and dispense liquid. Their measurement accuracy varies greatly depending on the style.

Types of pipette

i. Micro pipette

a. Micropipettes are a type of adjustable micropipette that delivers a measured volume of liquid.
b. Depending on size, it could be between about 0.1 µl to 1000 µl (1 ml).
c. Micro pipettes are used when accuracy is mandatory.
d. These pipettes require disposable tips that come in contact with the fluid.

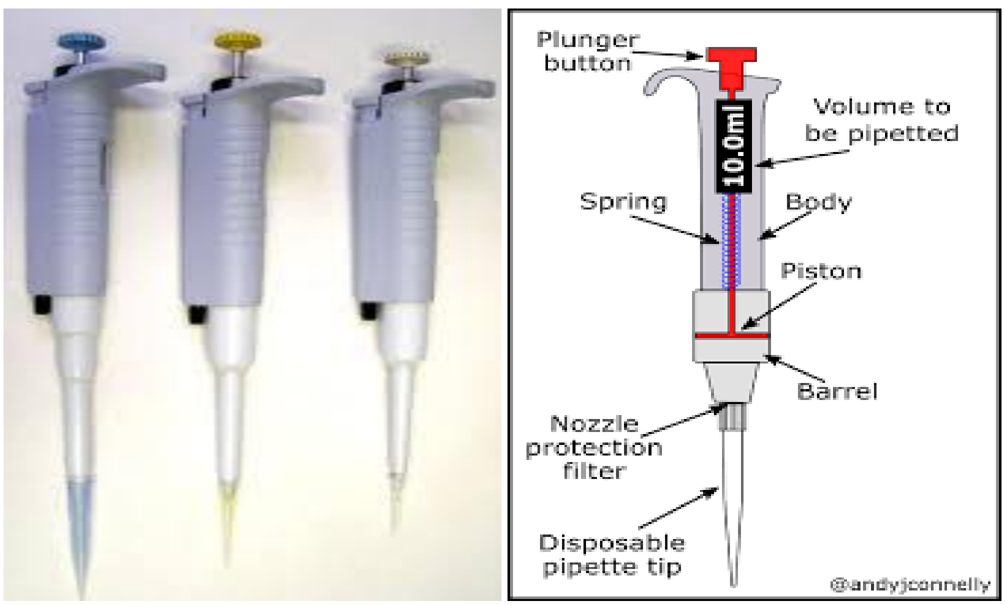


Figure 1.29: Micro pipit

Source ;(https://www.amazon.com/Microlit-Lab-Micropipette-Single-Channel-Autoclavable/dp/B07C1DW8JX(accesses on 29/08/2022)

ii. Graduate pipette

a. Graduated pipettes are a type of macro-pipette consisting of a long tube with a series of graduations, to indicate different calibrated volumes.

b. They require a source of vacuum at the bottom end.

Figure 1.30. Graduating pipette

Source;(https://www.amazon.com/Graduated-Pipette/s?k=Graduated+Pipette(29/08/2022)

- **Test tube**
  - ✓ A test tube is a clear glass or plastic container that is much longer than it is wide, commonly has a U-shaped bottom, and has an open top.
  - ✓ Test tubes are used to hold, mix and heat chemical.
  - ✓ They are used to culture (grow) microorganisms; collect samples as well.
  - ✓ There are several material **types of test tubes**: glass, plastic, metal and ceramic.
  - ✓ Glass and plastic are the most common while metal and ceramic are less common.

- **Test tube rack**
  - ✓ The **test tube rack** is an apparatus used in a laboratory to hold and transport **test tubes** during experiments or while examining cultures.
  - ✓ They can also accommodate other lab tools, such as stirring rods and pipettes. **Test tube racks** may also be referred to as **test tube** holders.

Figure 1.31: Test tube and Test tube rack

Source : https://www.eiscolabs.com/products/ch0004c 29/08/2022)

- **Inoculating Loop**
  - ✓ An **inoculation loop**, also called a smear **loop**, **inoculation** wand or micro-streaker, is a simple tool used mainly by microbiologists to retrieve an inoculum from a culture of microorganisms.
  - ✓ Heat fixation allows the bacteria to stick to the slide, instead of rinsing off with the various rinses used during the staining process.
  - ✓ Sterlizing of the **inoculation loop** is essential in order to avoid contamination of the slide preparations and also to prevent contamination of the inoculum plate.

Figure 1.32: Inoculation loop

Source ;(https://www.amazon.com/A2Z-IL05I-A2ZSCILAB-Bacterial-Inoculating-Insulated/dp/B07WKTTL5N29/08/2022)

- **Pestle and mortar**
  - ✓ **Mortar and pestle are**: A **mortar** is a vessel in which substances are ground or crushed with a **pestle** and it is typically made of hard wood, metal, ceramic, or hard stone, such as granite.
  - ✓ A **pestle** is a tool **used** to crush, mash or grind materials in a **mortar**.

Figer:1.33: Mortar and pestil

Source ;(https://www.homesciencetools.com/product/mortar-and-pestle-500-ml/ 29/08/2022)

C. **Tools and equipment used for disease control and prevention**

- **Tools ad equipments used for disease control**
- **Treatment syringe and needle**

Figure 1.34: Treatment syringe and needle

Source: (https://www.alibaba.com/product-detail/animal-needles-16G-full-size-veterinary_60493033930.html 30/08/2022)

- **IV Catheter**
  - ✓ Animals require IV, or intravenous, catheters when dehydrated, sick or about to be anesthetized.
  - ✓ Vet techs must place these small tubes in one of the animal's veins to allow for administration of liquids and medications.
  - ✓ Catheters enable rapid distribution of fluids throughout the patient's body.

Figure 1.35:- picture of IV catheters

D. Surgical tools and equipments

- **Emasculator:-** Used to daraw out, cut as well as seal the spermatic cordin an open castration.

Figure 1.36:- Emasculator

Sorce (https://www.coburn.com/12-hausmann-emasculator acesses on 27/08/2022))

- **Scalpel handle and blade: -** Used to incise tissues and cause minimal trauma to penetrate the desired depth in surgical operation (minor or general surgery).

Figure 1.37: Scalpel handle and blade

Source (https://veteriankey.com/surgical-instrumentation/(accesses on 27/08/2022))

- **Scissors; -** Used to cut tissues and suture materials except wire in surgery.
- **Retractors: -** used to facilitate exposure of the surgical field with little trauma.

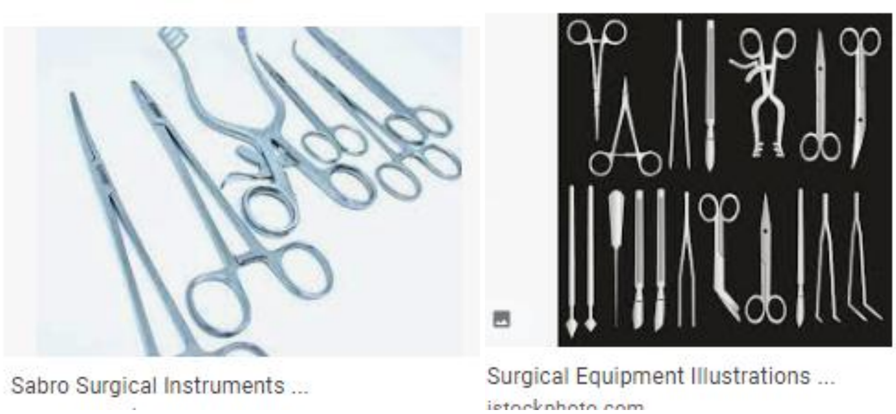


Figure 1.38: Different types of scissors and retractors

Source (https://www.istockphoto.com/illustrations/surgical-equipment(aceses on 27/08/2022))

- **Suturing material:** used to hold tissues in apposition until the wound can withstand outside stresses. plays an important role in wound repair by providing homeostasis and support to the healing tissue.

Figure 1.39. Suturing material

- **Surgical needle holder: -** Grasps and manipulate suture needles through the tissues
- **Tissue forceps:** Used to catch and hold tissues to facilitate deeper exposure of tissues or facilitating suturing.

Figure 1.40.Surgical needle holder, scissors and Tissue forceps

Source:(https://www.google.com/search?q=surgical+equipments&sxsrf=ALiCzsaBuZ_zSLFTid8_5QIwIejO3JH7LA:1661697315906&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjKg_Tx4On5AhWHh_0HHd60C8EQ_AUoAXoECAEQAw&biw=1366&bih=657&dpr=1(acesse on 27/08/2022))

- **Trocar and Cannula** Used to treat secondary bloat in large and small animal.

Figure 1.41. Trocar and cannula

Source (https://vetmeddevices.com/product/cattle-trocar-with-cannula-10x150-mm/) accesses on 27/08/2022)

- **Burdizzo castrator:-** Is an instrument used to crush spermatic cord through the skin for bloodless castration in ruminants.

figure 1.42. burdizo

https://www.exportersindia.com/product-detail/stainless-steel-burdizzo-castrator-5861974.htm 30/08/2022

- **Nap-sack:** Uses of sprayer
    - ✓ Application of acaricides in order to kill weeds
    - ✓ Application of insecticides to control various insects and pests
    - ✓ Application of fungicides to minimize the effect of fungal diseases.

Figure 1.43. sprayer

Source; https://chinagros.en.made-in-china.com/product/KqtmBeREOsUx/China-Knapsack-Sprayer-Hand-Sprayer-Manual-Sprayer-Backpack-Sprayer-Matabi-Sprayer-Agros-Sprayer-AM-S18-.html

Naso-gastric tube: used to dreanch medicine to equine

Figure 1.44 Naso gastric tube

Source: https://www.polymedicure.com/nasogastric-feeding-tube/ 30/08/2022

- **Boling gun**: Long metal or plastic instrument with a cup-shaped depression at one end. Used for placing solid medicine in the posterior part of the mouth of a horse or ox.

Figure 1.45. boling gun

- **Drenching gun**: Used to give medicine to animal in liquid form

Figure 1.46. Drenching gun

E. Tools and Equipment used for Husbandry Practices

- **Burdizzo :** Used for blood less castration of cattle, sheep and goat

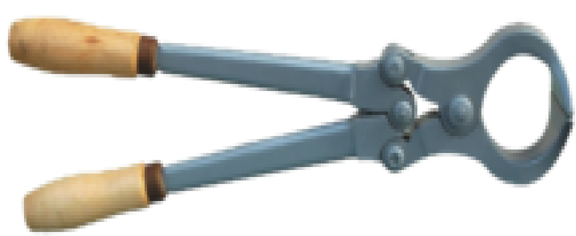


Figure 1.47: burdizo

- **Dehorning wire and saw:** Dehorning wire is made up of metal wires used to remove over grown horn of livestock.
  - ✓ **Dehorning** is the process of removing the fully grown horns of livestock (Cattle, sheep, and goats) for economic and safety reasons.
  - ✓ Commonly recommend dehorning young calves to: reduce the risk of injury and bruising to herd mates.
  - ✓ Prevent financial losses from trimming damaged carcasses caused by horned feedlot cattle during transport to slaughter.

Figure 1.48: Dihorning saw

- **Hoof Rasper/nipper:** Hoof nippers are used to trim the frog of the hoof.

Figure 1.49: Hoof Rasper

- **Hoof knife:** Hoof trimming is good for hoof health, and hoof knives are one of the most essential tools to use for this purpose.
- **Hoof trimmer**
  - ✓ The trimmer is used to remove large pieces of over grown hoof so that the smaller instruments can be used to shape the hoof.

Figure 1.50; Hoof trimmer and Hoof knife

Source : (https://www.alibaba.com/product-detail/Metal-Steel-Long-Handle-Horse-Hoof_60390571344.html 30/8/2022

F. Tools and Equipments used for disease prevention

- **Needle and Syringe**
  - ✓ Vet techs often complete tasks that call for the use of a needle and syringe.
  - ✓ A needle, used in conjunction with a syringe, injects or withdraws liquids.
  - ✓ A syringe, a glass or plastic cylinder marked with units of measurement, contains a piston system that enables specimen collection. Some of the more common tasks vet techs complete with a needle and syringe include drawing blood, injecting patients with medication or vaccines and performing needle

Figure 1.51:- Vaccination syringe

source: (https://zootecnicainternational.com/featured/injection-syringes-and-needles-for-the-animal-production-industry/ 30/8/2022)

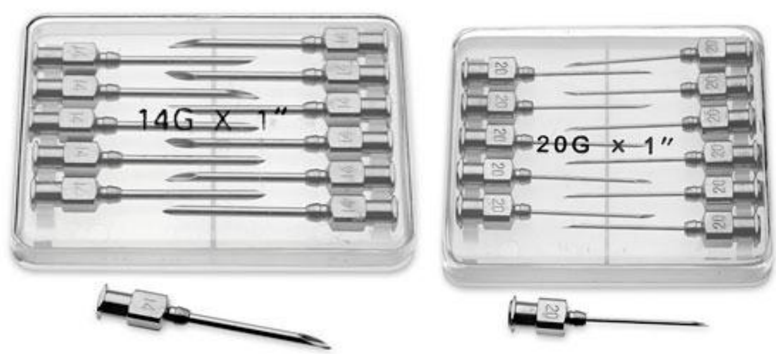


Figure 1.52:- picture of needle used to give vaccine

- **Ice box and Ice pack :** used to maintain cold chain for vaccine storage and distribution

Working of an Ice Cooler Box: Basics ...
medium.com

Ice packs - Standard products catalog...
itemscatalogue.redcross.int

Figure. 1.53: Ice box and Ice packe

Source ;(https://itemscatalogue.redcross.int/health--3/cold-chain--3/various-cold-chain-materials--119/ice-packs--XCOLICEP.aspx(30/8/2022)

1.1.3. Restraining tools and equipments

Restraint and approach to animals Restraint is the term used to imply control of an animal and may be necessary for medical reasons and nonmedical procedures. Animals are often resisting to the clinical examination procedures. The animal must be restrained so that it can be examined carefully, safely and with confidence the methods of restraint should be done, in order to able to carry out the examination safety and without danger to the clinician and assistants, the methods available may be classified as the following.

- Physical restraint.
- Chemical restraint
- Verbal/Moral restraining

Figure 1.54. Nose ring with rope and bull holder.

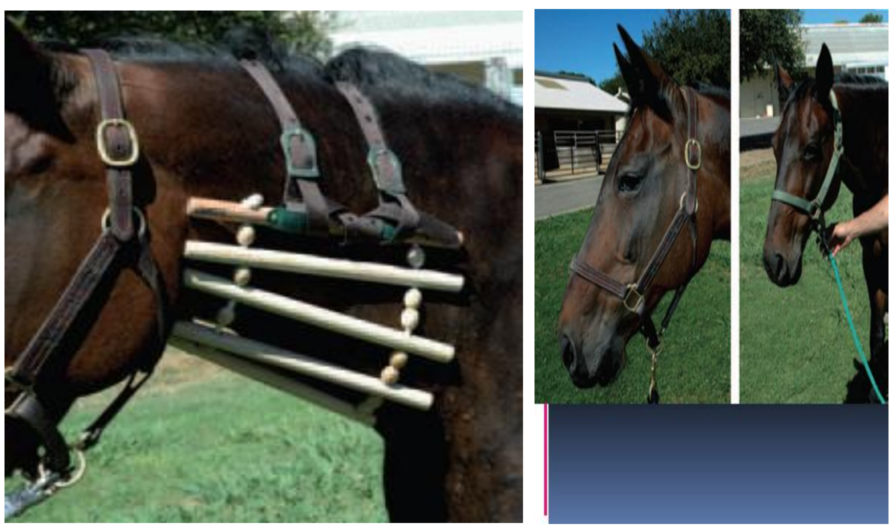


Figure1.55. Cradle and halter with lid

Figure 1.56. Nose twitch

https://www.google.com/search?q=tools+used+for+restraing+of+animals+&tbm=isch&ved=2ahUKEwjxxrCswOb5AhWUNewKHfd0CU0Q2cCegQIABAA&oq=tools+used+for+restraing+of+animals+&gs_lcp=CgNpbWcQAzIFCAAQgAQyBQgAEIAEMgUIABCABDIFCAAQgAQyBQgAEIAEMgUIABCABDIFCAAQgAQyBQgAEIAEMgUIABCABDIFCAAQgAQ6BAgAEENQtARYm19g4GZoAXAAeAGAAfQOiAHuOJIBBTctMi4zmAEAoAEBqgELZ3dzLXdpei1pbWewAQDAAQE&sclient=img&ei=XMgJY_GoOZTrsAf36aXoBA&bih=657&biw=1366(accesses on 26/08/2022)

Figure 1.58. Hobbles and milker's restraint

Figure 1.59:. Crutch and Home constructed camel chute

Figure 1.60. Dog collar &chain and muzzle

Source: https://www.shutterstock.com/image-photo/pitbull-terrier-muzzle-on-leash-1081550900 (acessed on 29/08/2022)

1.1.4. Tools and equipments used for Obstetrics and gynecological tools and equipments

- **Fetotomy wire**:- used to cut parts of fetus with in uterus
- **Fetotomy knife:-** used to cut parts of fetus with in uterus
- **Obstetric hook (blunt & sharp):-** used to hold nose and legs of fetus
- **Obstetric fetus repeller:-**
- **Hooks:-** used to hold nose and legs of fetus
- **Vaginal speculum:-** used to open the vagina to examine internal wall.
- **Teat cannula:-** used to treat mastitis
- **Teat slitter:-** used to open blocked
- **Teat dilator:** to remove blockage of teat



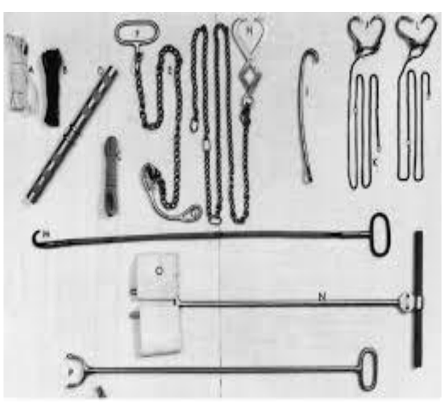



Figure1.61.foetotomy wire, hooks, vaginal speculum, fetotomy knife, obstetric hook (blunt & sharp), obstetric fetus repelled, teat slitter, teat bestiary, teat canula respectively

https://www.youtube.com/watch?v=xkx6wIFqmP0, accesses on 25/08/2022

source: https://depositphotos.com/stock-photos/gynecologist-instruments.html(date on 29/08/2022)

1.1.5. Artificial insemination instruments

- **AI sheath:-** used to cover the AI gun to cover and prevent the strew not remain inside uterus
- Nitr**ogen (jar) container:-**it help to hold liquid nitrogen
- **AI gun:-** Used to deposit semen
- **Scissors:** Used to cut the tip of non cotton parts of the straw
- **Thermometers:** It is used to measure the temperature of the water
- **Forceps**: the larger forceps used to hold and pickup semen straw from canister.

Figure1.62. AI sheath, glove, nitrogen container and AI gun respectively

https://www.vet-ebooks.com/list-of-veterinary-equipment-and-tools(accesses date 24/08/2022)

1.1.6. Tools and equipment used for cleaning activities. (add photo)

- **Cleaning** is removal of dirty and debries from tools , equipments and working environment
- **Disinfection** is a process of killing all the microorganism but not its spore
- **Sterilization** is a process of killing all the microorganism but its spore

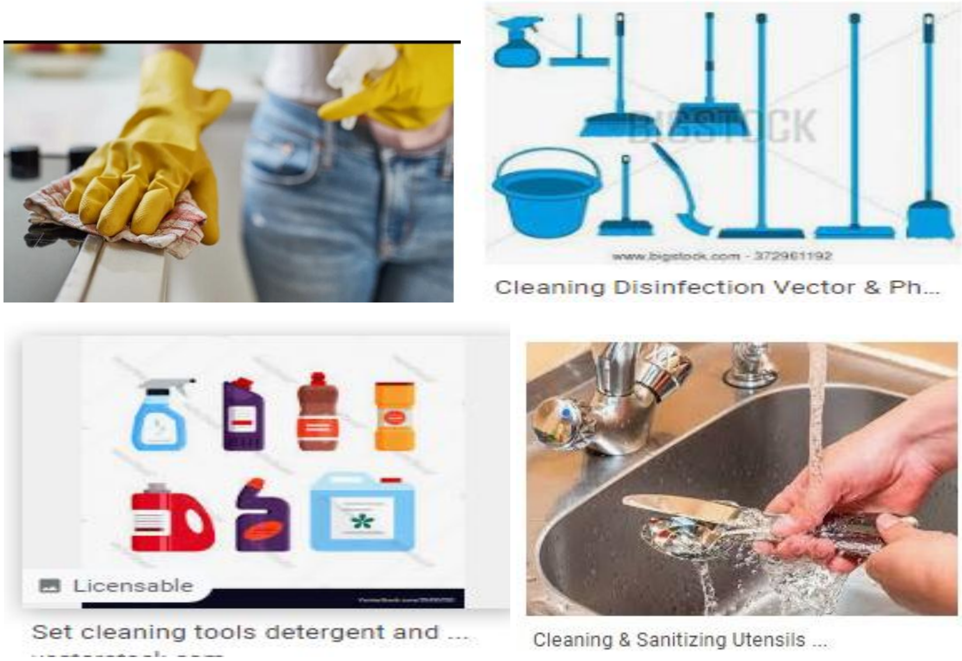


Figure 1.62: Show tools used for cleaning and disinfection (Rube, Glove, Bucket, Mop, Brush, and different Detergents)

https://cleaning.lovetoknow.com/cleaning-tips/correct-steps-cleaning-sanitizing-utensils-by-hand(accessed date on 27/08/ 2022)

https://www.bigstockphoto.com/image-372961192/stock-vector-cleaning-and-disinfection-tools-and-products-clean-floor%2C-vector-sanitary-and-hygiene-products%2C-bro(accessed date on 27/08/ 2022)

1.1.7. Tools used for waste disposal

Healthcare facilities generate different kinds of waste of which medical waste is just one. The facilities that fall into this category include blood banks, hospitals, veterinary clinics, physicians' offices, laboratories, medical research facilities, and dental practices. Medical waste can be defined as healthcare waste that could be contaminated by body fluids, blood as well as other materials that could potentially be infectious. Regulated medical waste is the name by which this type of waste commonly goes. According to the Practice Green health Sustainability Benchmark

Report 2015, one regular operating room results in 5.4 tons of waste every year, which costs the hospital $5.243.

Health Care Waste Management – Public ...

Trash Bucket for Used Medical Mask ...

waste disposal | Types & Methods ...

Figer : 1.64: different waste bucket and waste treatment plant

https://www.dreamstime.com/red-sign-trash-bucket-used-medical-mask-gloves-concept-icon-red-rectangle-sign-biohazard-trash-garbage-safely-dispose-image199320820

1.2.Pre-operational checks of basic veterinary tools and equipment

In order to ensure that your machine is operating safely and efficiently, a proper pre-operational check is important. The pre-operational check covers a number of key components of the machine and helps to identify maintenance issues and repairs that may be required. It is important to assess the conditions observed during your inspection to determine the action required. Appropriate responses could range from immediate removal from service, often maintenance or basic repair that you will be expected to perform, or reporting conditions that will require service later. check or pre operational inspection of equipments prior to every use will reduce the chance of equipment being operated in an unsafe condition. This makes easier to spot and deal with maintenance issues early before any turn in to problem causing downtime, equipment damage or expensive repairs. Unsafe machinery can also cause injury to the operator or other workers and damage to the facilities or product. Pre operational checks of equipments may include:-

- Cleaning,
- Lubricating
- Identifying and segregating unsafe or faulty equipment for repair or replacement
- If any equipment is to be checked or repaired, it should always be turned off and isolated so it can't be started in error.
- Most equipment now comes with guidelines for maintenance, including advice on how to carry out equipment checks safely.

1.3.Performing simulation trial to operate basic tools and equipment

- **Operation of tools and equipments**

The tools and equipment in veterinary medicine are different to undergo different procedure. That is, every tools and equipment has their own operation procedure when the tools do not operate as they are supposed to be, the manufacturing stage must be stopped immediately. This issue can be prevented with regular check-upsets ensure that you are not missing out on anything; you need to keep a record of every part being inspected. It also ensures that you record everything that needs further checkups in the future.

An excellent record for manufacturing inspection tends to fulfill the following criteria:

- Each component is labeled for ease of identification.
- Categorization of components for easier finding.
- Notes on past damages and repairs of the components.
- Staying up to date with the record.

- **Maintenance of tools and equipments**

Instruments are a major asset and represent a significant share of the total capital spending of veterinarian hospital and practice. The practical experience recorded in this guide, together with a description of fundamental interrelationship, is intended to help users to keep their reusable instruments in good working order and preserve their value for many years by ensuring proper care and maintenance. It should be emphasized that the recommended measure must always be carried out in according with the manufacturer's instruction, pertinent hygiene requirement and official safety at work guidelines.

A maintenance schedule should be in place to ensure that your equipment is maintained at least at intervals indicated in the manufacturer's operating instructions or more frequently if indicated by the risk assessment. Any daily checks should be undertaken as recommended by the manufacturer. This will help prevent problems such as blockages, leaks or breakdowns, which can increase risks

Regular maintenance is essential to keep equipment, machines and the work environment safe and reliable. Lack of maintenance or inadequate maintenance can lead to dangerous situations, accidents and health problems. Maintenance is a high-risk activity with some of the hazards resulting from the nature of the work. Maintenance is carried out in all sectors and all workplaces. Therefore, maintenance workers are more likely than other employees to be exposed to various hazards

Maintenance is a generic term for variety of tasks in very different types of sectors and all kinds of working environments. Maintenance activities include:

- inspection
- testing
- measurement
- replacement
- adjustment
- repair
- upkeep
- fault detection

- replacement of parts
- servicing
- Lubrication
- cleaning

1.4. Identify, Segregate, and Report Unsafe or Non-Functional Tools and Equipment

The practice of veterinary medicine involves the risk of physical injury through contact with animals and equipment, repetitive motion, and motor vehicle accidents. In a German study, the rate of accidents reported to an insurance database was 2.9 times higher for veterinarians than for physicians in general practice. Unsafe machinery can also cause injury to the operator or other workers and damage to the facilities or product. . As a result, there might be issues in the quality of manufactured products. There is also an increased risk of accidents when manufacturing with defective equipment.

1.4.1. Inspection and Identification of Unsafe Tools and Equipments

Inspection is done to detect defects on any component in the manufacturing process. When a tool, material, or equipment is defective, the manufacturing process can be disrupted. You should inspect work equipment if your risk assessment identifies any significant risk (for example, of major injury) to operators and others from the equipment's installation or use. The result of the inspection should be recorded and this record should be kept at least until the next inspection of that equipment. This will depend on type of work equipment, its use and the conditions to which it is exposed. This should be determined through risk assessment and take full account of any manufacturer's recommendations

In what condition does a material become considered as unsafe?

Materials are designated as "hazardous materials" when they pose a significant risk to people or property. There are two primary ways that a waste material can become classified as a hazardous waste,:

- **Listed wastes:** Wastes from certain industrial processes are automatically classified as hazardous. Each waste of this type is given a code number. The full list of hazardous waste codes appears in the Code of Federal Regulations,40 CFR 261.
- **Characteristic wastes**: Wastes that do not appear on the CFR lists may nevertheless be classified as hazardous if they have one of four properties:
  - ✓ Ignitability
  - ✓ Corrosivity
  - ✓ Reactivity
  - ✓ Toxicity

1.4.2. Reporting, Record Keeping, and Quality Assurance

Given that workers who have access to animals used in research are required to receive medical support services, the medical provider must have a mechanism for communicating who has received the pre-placement medical evaluation and any required periodic medical evaluations (e.g., testing workers with access to non-human primates for recent infection with *M. tuberculosis*). As mentioned above, injury reports provide valuable information. The health care provider promptly should share the circumstances of work-related injuries with their counterparts in safety and management. This approach permits a timely review of incidents and ideally the identification of potential corrective actions that may prevent recurrences. The data is also needed for generating an OSHA 300 log. The health care provider should be familiar with the workers' compensation application process and facilitate workers' completion of the reporting forms. Reviewing injury data for a specified period may help identify critical patterns in the injury data (e.g., a disproportionate share of the injuries involves a particular process or piece of equipment).

1.5. Understanding of Occupational health and safety (OHS)

Occupational Health and Safety (OHS) is about working safely and ensuring that the work place is a safe place for everyone, including visitors to the Property. It is normal to think that 'I'm Ok, nothing will happen to me.' But if we are careless around the workplace then accidents will happen. The results may be minor like a cut finger or bruised leg. However, when you think about the machinery, chemicals and other hazards on a in livestock farm, you realize that it could be a very dangerous place.

1.6. Occurrence of Work place hazard and its identification.

Like any other jobs, there are professional hazards, which occur at veterinary laboratory while performing laboratory activities. Veterinary occupational health and safety policy and procedure help to reduce these veterinary occupational risks. Veterinarians, animal handlers, animal health technicians and animal health assistances can be exposed to occupational health risks, In the work place workers will always be exposed directly or indirectly to hazards.

1.6.1. Identification of work place hazards in veterinary are

A. **Physical and mechanical hazards**- are common sources of injuries and include

- Falls(especially in health care , transport and construction)
- Confined spaces
- Noise
- Temperature(cause heat stroke, exhaustion, cramps, rashes, dizziness
- Risk of burns- dehydration
- Vibrating machinery
- Cold stress-hypothermia
- Lighting
- Electrical injury
- Air pressure

B. Biological hazards

: Biological hazards are: hazard immerged from pathogen or infectious agent from working area

- Bacteria (tuberculosis)
- Virus
- Blood born pathogens
- Fungi (Mold

C. Chemical hazards

- Heavy metals
- Solvents
- Petroleum
- Fumes(noxious gases/vapor's)
- Heavily reactive chemicals
- Fire, conflagration and explosion hazards
- Explosion
- Deflagration
- Detonation
- conflagration

D. psychosocial hazards

- ✓ precarious work contracts
- ✓ increased worker vulnerability due to globalization
- ✓ new forms of employment contracts
- ✓ feeling of job insecurity
- ✓ aging work force
- ✓ long working hours
- ✓ work intensification
- ✓ lean production and out sourcing
- ✓ high emotional demands
- ✓ Poor work life balanc

E. **Ergonomic hazard**(repetition, forceful exertions, awkward postures, contact stress, vibration, work area design and tool or equipment design) Therefore, it is necessary to have exposure control plan trying to minimize the risk.

1.6.2. Maintaining safe work place

Safe environment /work place are the area where professional risk is controlled or minimized to harmless level. Veterinarian and animal health technician have to have the knowledge of what causes injuries & disease, is easier to design and implement suitable measures to wards privations certain safety rules must be followed when attempting to capture, restrain, treat, feed or exercise animals. Accident could happen just anywhere. It could happen while you were just walking out on the streets or same place else. But, it is especially common in workplaces. However, there are safety precautions that can be undertaken to minimize the risk. There’s this thing we call safety hazard. Safety hazards may be defined as a discipline aimed at protecting the individuals from risks and impending danger. This should be highly stressed especially on workers to prevent any injury and accident.

- **This article tackles about some of the safety hazards that must be observed especially at work areas are:**
  - ✓ Before directly joining the veterinary clinical activity, one must ensure the safety of work place. The animals, which are going to be presented to theatre, should be handled and restrained carefully.
  - ✓ Restraining equipments and any material used in the veterinary clinic should be checked for functional status.
  - ✓ Any materials which is damaged or non functional should get maintenance before work operation.
  - ✓ Veterinary clinic environment should be clean and free of any dirty materials that may cause health problem in animals as well as workers.
  - ✓ Waste materials that come from animals should be disposed off to avoid slipping. Slippery steps or floor that may cause falls to animals must be corrected.
  - ✓ Electric shock is a possible danger in damp /wet areas. Electrical cords that may come into contact with water are hazards. Water that over flows or faucets that do not turn off completely may cause slippery conditions or electrical shock hazards.
  - ✓ To reduce exposure to a livestock incident or illness
    - understand animal behavior
    - provide proper and safe facilities
    - protect against zoonotic diseases
    - Wear personal protective equipment.

Self-check 1	Written test

Name…………………………………………… ID………………………… Date……

Directions: Answer all the questions listed below.

Test I: Choose the best answer (each has 1 point)

1. Which machine is used for sterilization of tools and equipment which can with stand dry high temperature 160^{o}c for 2hrs?
   A, water bath B, Autoclave C, Hot Air Oven D, None
2. Which tools is used to culture or facilitate growth of cell or microorganism under required condition of temperature, humidity and corbon dioxide?
   A, Autoclave B, Incubators C. Hot Air Oven D. Water bath
3. Among the following which is **NOT** a diagnosis instrument?
   A, thermometer B, stethoscope C, ultrasound D, bolng gun
4. Which one is different from the other? :
   A, suture material B, hemostatic forceps
   C, scalpel handle and bled D, vaginal speculem
5. Among the following which one is not an example of treatment instruments?
   A. Syringe B. Cather C. boling gun D. Thermometer
6. Among the following which one is not a surgical tools and equipments?
   A, Scissor B Scalpel handle C Bled D, Stomach tube
7. **Which one is not a biological hazard ?**
   A, bacteria B virus C fungs D, heavy metal.

Test II: Short Answer Questions (each has 4 pt)

1. Write the role or function of the following AI instruments? (2 pt)
   A. Thermometer B, AI gun C, scissor D, Forceps E, liquid nitrogen.
2. Write the different type of syringes based on their function?
3. Write the two type of microscope based on their power source?
4. List at least two instrument used for **diagnosis** purpose?
5. Write the **restraining method** and **material** used for the following specious of animals
6. Write the type of waste
7. What is OHS
8. List the type of work place hazard

Note: **Satisfactory rating - 25 points Unsatisfactory - below 25 points**

Operation Sheet -1

1.1 Techniques/Procedures to wear and remove PPE

A. Tools and material required

- ✓ Distilled water
- ✓ Detergent/soap
- ✓ Different PPE (gloves, mask or respirator, eye goggle and gown)

B. Procedure

- ✓ first check the presence of appropriate PPE
- ✓ wash and clean your hand with soap and water
- ✓ Step to wear PPE
  - first wear gown, ⇨ mask or Respirator , ⇨ eye goggle or face shield ⇨ Glove
- ✓ To remove PPE, follow the following steps
  - First remove glove, ⇨ eye goggle, ⇨ Gown, ⇨ Mask or respirator ⇨ then wash hand

Source ((https://www.youtube.com/watch?v=quwzg7Vixsw accessed date on 25/08/2022)

1.2. Procedures or techniques to handle and operate microscope

A. Tools and equipment used

- Microscope with it accessory
- Electric power source and socket
- Cleaning tools (cotton, gauze, and xyeline)
- Oil emersion
- Smear or specimen
- PPE(glove and gown)

B. Procedure

Step 1. To carry the microscope grasp the microscopes arm with one hand. Place your other hand under the base

Step 2. Place the microscope on a table with the arm toward you.

Step 3. Turn the coarse adjustment knob to raise the body tube

Step 4. Revolve the nosepiece until the low-power objective lens clicks into place.

Step 5. The diaphragm. While looking through the eyepiece, also adjust the mirror until you see a bright white circle of light.

Step 6. Place a slide on the stage. Center the specimen over the opening on the stage. Use the stage clips to hold the slide in place.

Step 7. Look at the stage from the side. Carefully turn the coarse adjustment knob to lower the body tube until the low power objective almost touches the slide.

Step 8. Looking through the eyepiece, very slowly the coarse adjustment knob until the specimen comes into focus.

Step 9. To switch to the high power objective lens, look at the microscope from the side. Carefully revolve the nosepiece until the high-power objective lens clicks into place. Make sure the lens does not hit the slide.

Step 10. Looking through the eyepiece, turn the fine adjustment knob until the specimen comes into focus.

1.3. Procedures to operate and use Autoclave

A. Tools and equipments

i. Water
ii. Electric source
iii. Autoclave machine
iv. Material needs to be steralized

B. Procedures:

1. Put a little water in the bottom of the pressure cooker.
2. Prepare the material to be sterilize
3. Load the material to be sterilized into the autoclave.
4. Screw down the lid. 5. Open the steam valve.
5. Switch on. If there are high and low switches on the autoclave make sure both are switched on.
6. Let steam come out for at least five (5) minutes before closing steam valve. Continue heating until the pressure is up to 15 psi.
7. Adjust pressure and turn the heat down or the high switch off.
8. Leave to steam for the appropriate time then turn off the autoclave.
9. Leave to cool to reduce the pressure to zero.
10. Open the steam valve to release any remaining pressure.

11. Wait five (5) minutes before opening the lid.
12. Open the lid and unload the material in the autoclave

1.4. Steps to light and adjust a Bunsen burner

A. Tools and equipments

I. Matches' or lighter
II. Cylinder containing gasses
III. Cable wire or burner hose

B. Procedures:

Step 1. Connect the burner hose to the gas outlet. The gas outlet handle should be in the fully closed position with the handle at a right angle to the outlet pipe.

Step 2.Turn the gas valve on the gas outlet to the fully open position (handle is now parallel to gas outlet).

Step 3.With the main gas valve fully open; open the gas adjustment at the base of the burner by several turns to give

Step 4.Light the burner by holding a match to the side of the mouth of the burner. If you stick the match in the middle of the gas stream, the flame is usually blown out before the burner lights

Step 5.Adjust the amount of air in the flame by turning the barrel of the burner. A flame with too little air mixed in it has a light blue, bushy appearance (see left) and more air must be mixed in by unscrewing the barrel to the left.

1.5. Procedure to operate and use Hot air oven

A. Tools and equipments

i. Electric source and cables
ii. Hot air oven machine
iii. Material to be sterilized

B. Procedures:

Step 1. First arrange the material to be sterilized loosely and evenly on the racks of the oven allowing free circulation of air and thereby even heating of the load.

Step 2. If desired, warp instruments by aluminum foil or place in a metal container or by trays

Step 3. Do not pack the load tightly since air is a poor conductor of heat.

Step 4. Switch on the power supply and control the temperature of the oven by adjusting the thermostat.

Step 5. Note the time when the desired temperature is reached (heating-up time).

Step 6. Hold the load in the oven at this temperature for a definite period of time (holding period). This is usually 60 minutes at 160oC

Step 7. Do not overheat since it would char the cotton plugs and paper wrappings.

Step 8. Cool down the oven

Step 9. After cooling remove packs or metal containers and store.

1.6. Steps to Operation of water path

A. Tools and equipments

i. Water bath machine
ii. Electric source, and cable
iii. Material, need to be sterilized or bathed

B, Procedures:

Step 1. Set up on a stable surface, away from flammable and combustible materials including wood and paper

Step 2. Put a little water in the bottom of the water bath.

Step 3. Prepare the material to be sterilize

Step 4. Load the material to be sterilized into the water bath.

Step 5. Switch on water bath and heat for a given period of time

Step 6. Turn off the heat down or the high switch off.

Step 7. Leave to cool to reduce heat of water bath.

Step 8. Unload the material in the water bath

1.7. Techniques to identify the type of thermometer and its operation

- **Tools and equipments**

i. Thermometer (both mercury/Analog and Digital)
ii. Alcohol (disinfectants)
iii. Cotton or gauze
iv. Lubricants (paraffin oil)

- **Procedures:**

Step 1. The thermometer should be sterilized by disinfectant (antiseptics) before use;

Step. 2. It should be well shaken before recording of temperature to bring the mercury column down below the lowest point likely to be observed in different species of animals.

Step 2. The bulb end of the thermometer should be lubricated with liquid paraffin or glycerin or soap especially in case of small pup and kitten.

Step 3. Care should be taken so that the bulb of the thermometer remains in contact with the rectal mucous membrane.

Step 4. The thermometer should be kept in site for at least 3-5 minutes.

Step 5. Read the thermometer

NB. How to operate digital thermometer (https://www.youtube.com/watch?v=XuL0K9kxu3s (Accesses on 27/08/2022)

LAP TEST-1	**Performance Test**

Name……………………………… ID…… Date………………

Time started: ________________________ Time finished: ________________

Instructions:

- To perform wearing PPE and cleaning tools, considerate to check presence of all required facility which is listed in information sheet and operation sheet.
- Identify and check the presence of required or necessary tools and equipment that enable you to perform the task you want to perform.

Task-1 perform how to wear and remove or take off PPE

Task 2. Handle and operate the microscope

Task 3. Perform how the Autoclaves work

Task 4. Adjust and use Bunsen burner

Task 5. Operate Hot air oven

Task 6. Check and use the water bath

Task 7. Check and use the thermometer.

LG #2	LO #2- Undertake cleaning activities

Instruction sheet 2
This learning guide is developed to provide you the necessary information regarding the following content coverage and topics: • Cleaning of tools and equipment • Reporting of problems or difficulties in the work place • Disposing of waste material This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to: • Undertake Cleaning activity in a safe and environmentally appropriate manner according to organizational guidelines. • Report Problems or difficulties in completing work to the required standards or timelines to the respective hierarchy. • Store Waste material produced during cleaning activities in a designated area according to supervisors' instructions.
Learning Instructions:
1. Read the specific objectives of this Learning Guide. 2. Follow the instructions described below. 3. Read the information written in the information Sheets 4. Accomplish the Self-checks 5. Perform Operation Sheets 6. Do the "LAP test"

Information Sheet 2:

2.1.Cleaning, Disinfection and Sterilization of Tools and equipment

- **Cleaning:**

Proper instrument disinfection and sterilization clearly depend on adequate cleaning and care, as well as on selecting the right material and using suitable cleaning agents and treatment processes in order to maintain the value of the instruments.

The cleaning step should never be skipped because residues (e.g. from packing materials or care agents) could lead to the formation of stains or deposit during sterilization

- ✓ **Cleaning can be classified as**
  - Dry cleaning

Dry cleaning is a primary type of cleaning activity by using dry cloth, sponge, cotton or gauze to remove dirty material like dust, ash and any other dirty from tools and equipments etc

- Wet cleaning

Wet cleaning can be performed by using water and detergent chemical to remove greasy dirty material from tools and equipment used in veterinary service delivery system.

- ✓ **Cleaning, can be also classified as**
  - ✓ **Manual cleaning** :

For manual cleaning, active non protein fixing cleaners with or without antimicrobial effect or enzyme are used. If disinfecting cleaning is required, the disinfection capability should be proven under 'dirty condition (high protein load)

- ✓ **Machine cleaning**

Before machine cleaning, wet cleaning should be performed and items must be first be thoroughly rinsed. This is because foam impairs the cleaning and disinfection result in machine based processing.,

Disinfection

- **Disinfection:**
  - ✓ it is process of killing microorganism from any veterinary tools and equipment but not the spore
  - ✓ can be carried out either by thermal or chemical processes.
  - ✓ Thermal disinfection is preferred whenever possible. It is generally more reliable than chemical processes, leaves no residues, is more easily controlled, and is non-toxic.
  - ✓ Heat sensitive items have to be reprocessed with a chemical disinfectant. Organic matter (serum, blood, pus or fecal material) interferes with the antimicrobial efficiency of either method.
  - ✓ The larger the number of microbes present, the longer it takes to disinfect. Thus scrupulous cleaning before disinfection is of greatest importance.
  - ✓ There are three types of High Level Disinfection (HLD):
    - Disinfection by boiling
    - Moist heat at 70-100°C • Chemical disinfection

Note: When sterilization is not available, HLD is the only acceptable alternative for instruments and other items (=semi-critical items) that will come into contact with the bloodstream or tissues under the skin. Boiling is HLD, not sterilization. Flaming is not an effective method of HLD because it doesn‘t effectively kill all microorganisms.

- **Sterilization**:

Sterilization is defined as a process of complete elimination or destruction of all forms of microbial life (i.e., both vegetative and spore forms), which is carried out by various physical and chemical methods. Technically, there is reduction $\geq 10^6$ log colony forming units (CFU) of the most resistant spores achieved at the half-time of a regular cycle.

2.2. Reporting of problems or difficulties in the work place

Everyone wants a safe place to live and work. The challenge with livestock farm is that, both the livestock and their families live together and sharing the same house. Zoonotic disease can be transmitted easily. So that issues of animal safety includes (livestock safety policies) has to be passed as:

- The risk of being hurt physically by an animal that is frightened or has been startled.
- The risk of being hurt due to the misuse of equipment or equipment that is poorly maintained
- Developing allergies to animal hair, dander, or secretions such as saliva, urine, and secretions of various glands associated with the skin
- Diseases and microorganisms such as Salmonella, Campylobacter, Cryptosporidiosis, Q-fever, Brucellosis, Leptospirosis, enteric bacteria and parasites

Safe environment /work place are the area where professional risk is controlled or minimized to harmless level. Before directly joining the veterinary clinical activity, one must ensure the safety of work place like:

- The animals, which are going to be presented to the area, should be handled and restrained carefully.
- Restraining equipments and any material used in the veterinary clinic should be checked for functional status.
- Any materials which is damaged or non functional should get maintenance before work operation.
- Veterinary clinic environment should be clean and free of any dirty materials that may cause health problem in animals as well as workers.
- Waste materials that come from animals should be disposed off to avoid slipping. Slippery steps or floor that may cause falls to animals must be corrected.
- Electric shock is a possible danger in damp /wet areas. Electrical cords that may come into contact with water are hazards. Water that over flows or faucets that do not turn off completely may cause slippery conditions or electrical shock hazards.

Generally to reduce exposure to a livestock incident or illness

- understand animal behavior

- provide proper and safe facilities
- protect against zoonotic diseases
- Wear personal protective equipment

2.3. Identification and disposal of waste

2.3.1. Identification of waste

Waste is everything that no longer has a use or purpose and needs to be disposed of, The term certainly applies to discarded material, but there are specific definitions for waste that affect how waste is regulated and must be handled, especially in professional settings. The majority of household and veterinary practice waste is considered "solid waste," regardless of whether it's actually "solid" in physical form. The U.S. Environmental Protection Agency (EPA) *defines solid waste as* "any garbage or refuse, sludge from a waste water treatment plant, water supply treatment plant, or air pollution control facility and other discarded material, including solid, liquid, semi-solid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities**.**" Based on this definition, examples of solid waste generated by veterinary clinics include, but aren't limited to: animal tissues, fluids and carcasses; animal waste; chemicals used in laboratory, cleaning or radiographic procedures; syringes and other medical supply waste; some medications, such as epinephrine and nitroglycerin; chemotherapy agents; mercury from thermometers; light bulbs and batteries; pesticides; paint and other solvents; and more.

A common type of waste in veterinary are: classified as Hazardous and non hazardous:

- **Types of hazardous veterinary waste?**

✓ **Cytotoxic and cytostatic pharmaceuticals**

These are medicinal products that are toxic, carcinogenic, toxic for reproduction or mutagenic. They include glass bottles or vials, clinical items like swabs, gloves and masks, syringes and sharps and animal bedding.

✓ **Contaminated sharps**

Sharps should be subject to a separate risk assessment to determine the risk in their disposal. They may be contaminated with a substance that is a risk to human or environmental health, and therefore should always be disposed of properly. This category covers partially and fully discharged hypodermic needles, and other sharp instruments and objects.

- ✓ **Infectious, clinical waste**

Waste containing viable micro-organisms or other toxins is labeled as infectious, clinical waste. This waste must be disposed of correctly because it is reliably believed to cause disease in humans. Waste that is deemed a risk should also be classed in this category.

This category could include items used for treatment like swabs, bandages, masks and gloves, animal bedding and blood or body parts.

- ✓ **Chemical waste**

Chemical waste, such as sterilizers and disinfectants, photographic chemicals, disinfectant, and x-ray solutions may be dangerous to human consumption if they enter the water supply in high quantities, and so should be disposed of properly.

- **Types of Non-Hazardous Veterinary Waste?**

- ✓ **Offensive waste**

Veterinary non-hazardous waste is waste other than sharps that is not hazardous or clinical but may cause offense to the senses in some way. This waste will have undergone assessment that determined that it was not a risk to human or animal health.

Offensive waste could include swabs, masks and gloves, animal bedding and animal faces.

- ✓ **Sharps**

Sharps are any materials that have a pointed edge and they can cause damage to humans by piercing the skin. They must be subject to a risk assessment to determine that they pose no risk of infection to animals or people.

- ✓ **Pharmaceuticals**

Pharmaceuticals and other controlled drugs should be disposed of carefully as they may pose a risk to human or animal health if consumed.

This may include denatured controlled drugs, prescription medicines, out of date drugs and contaminated bottles, syringes, and packaging

- ✓ **Pet cadavers**

Pet cadavers are transferred and disposed of under animal by product controls, except when the cadaver is suspected of harbouring a disease

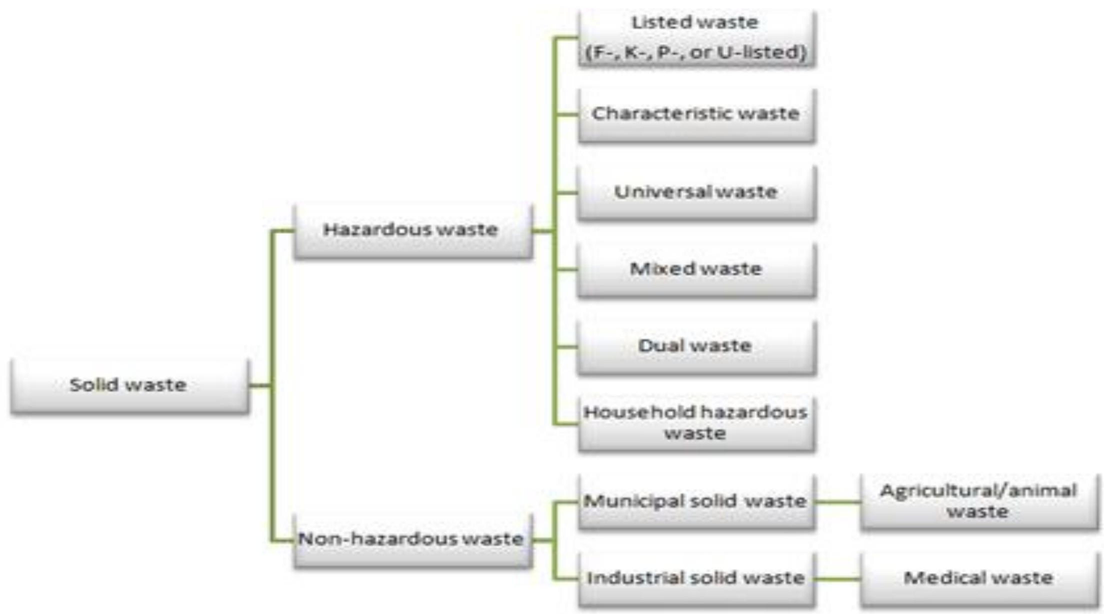


Figure. 2.1: Shows type and sourec of waste

Source: (https://www.dekomed.co.uk/vet/veterinary-equipment-what-you-need-when-setting-up-a-clinic/)

2.3.2. Disposal of waste material

Disposable instruments should never be used more than once as the respective declaration of conformity applies to single use only. Medical and hazardous waste generated during veterinary practice is regulated waste due to its potential to cause harm to anyone who comes into contact with it. For this reason, it is important and that anyone handling untreated medical or hazardous waste is provided with appropriate PPE to minimize risk. Employees are also required to have training in the proper use and disposal of PPE. Other types of training are also mandated for all employees who handle or are exposed to medical or hazardous waste. These include blood borne pathogens training, waste handling and classification regulations training, hazardous communications and chemical safety training, and DOT hazardous materials transportation regulations training

- **Handling and Disposing of veterinary hazardous waste**
  - ✓ ***Disposal of cytotoxic and cytostatic pharmaceuticals***

Any sharps should be segregated into sharps bins immediately after use while other cytotoxic and cytostatic waste, such as gloves and masks, should be disposed of in their relevant waste disposal bin.

- ✓ ***Disposing of contaminated sharps***

Contaminated sharps should be separated into yellow sharps containers immediately after use. Non-contaminated sharps can be further separated into orange-lidded bins.

- ✓ ***Disposing of infectious, clinical waste***

These items should be separated into the appropriate containers immediately and sent for high temperature incineration. Some items can also be further separated for suitable alternative treatment, like autoclaving.

- ✓ ***Disposal of photographic chemicals***

All liquid chemical waste, including those listed above, should be disposed of by being separated into leak proof containers immediately after use. The specific requirements for their disposal should be discussed with your waste removal provider.

- **Disposal of non hazardus waste**
  - ✓ ***Offensive waste disposal***

Offensive waste should be disposed of in landfill or in other suitable permitted facilities. Non-hazardous sharps might be an unused needle dropped on the floor that cannot be used, but if there is thought there could be a risk, however small, the sharp should be categorised as hazardous.

- ✓ ***Disposal of sharps***

Sharps should be disposed of immediately into the relevant sharps containers and disposed of by your waste provider.

- ✓ ***Disposal of pharmaceuticals***

All controlled drugs must be denatured or made not readily recoverable, and then disposed of with other pharmaceuticals. For Schedule 2 controlled drugs, this should be done in the presence of an authorised person.

- ✓ ***Disposal of pet cadavers***

Pet cadavers can be released for burial at home, burial in a pet cemetery or cremation.

Self-Check – 2	**Written Test**

Name………………………………………… ID………………………… Date…….

Directions: Answer all the questions listed below.

Test I: Multiple choices(1 point each)

1. Among the following which one is used to romove dirty from tools and equipments
   A, cleaning B, Sterilization C, Disinfection D, Boiling
2. Among the following which one is used to kill all the microorganism and its spore
   A, cleaning B, Sterilization C, Disinfection D, Boiling
3. Which one is not considered as a work place hazard
   A, Mis use of tools and equipment B, hairs and dander of animal
   C, clean space D, disease
4. Which one is not used to maintain safe work place
   A, Restraining C, improper handling of sharp material
   B, Cleaning D, Maintaining damaged tools
5. Among the following which one is used to reduce, incidence or illness on animals
   A, knowing animal behavior, B, Working with safe tools
   C, Transmiting zoonotic disease D, Wearing PPE
6. Among the following which one is not conciderd as waste
   A, Pet cadaver C, expired drug
   B, Sharp material D, functional tools and equipments

Test II: Short Answer Questions (2 point each)

1. Write the definition of cleaning, Disinfection & Sterilization :
2. List and explain the type of cleaning
3. List the tools and equipment used for disinfection and Sterilization
4. List methods of disposal

Note: **Satisfactory rating - 7 points** **Unsatisfactory - below 7 points**

You can ask you teacher for the copy of the correct answers.

Operation Sheet -2

2.1.Techniques or Procedures to clean surgical instrument and tools

A. Tools and equipments

- Surgical tools (scalpel bled, scalpel handle, scissors, forceps, etc)
- Washing dish
- Distilled water
- Brush
- Opsenic cleaning
- Detergent
- PPE
- Lubricants

B. Procedures/Steps/Techniques to clean veterinary tools and equipments

- ✓ First wear PPE
- ✓ Wash the washing dish
- ✓ Add distilled water on to washing dish
- ✓ Add detergent or soap. by measuring the amount
- ✓ Putt all the tools required to wash, in to washing dish contain soap and water
- ✓ After the tools and equipment moisten , brush it starting from the tip up to the shaft
- ✓ Next, put it in to pure distilled water, to rinse it
- ✓ Rinse items thoroughly with clean water to remove all detergent. Any detergent left on the items can reduce the effectiveness of further processing.
- ✓ Inspect items to confirm that they are clean.
- ✓ Allow items to air dry or dry them with a clean towel if chemical disinfection is going to be used. This is to avoid diluting the chemical solutions used after cleaning. Items that will be high-level disinfected by boiling or steaming do not need to be dried.

2.2. Steps or procedure to clean and disinfect different parts of microscope

A. Tools and equipments

- **Cleaning Solutions and Solvents**
  - ✓ Soap solution for cleaning of body and stage
  - ✓ Ether-Alcohol, Alcohol, or Lens Cleaner Solution for cleaning of lenses
  - ✓ Refer to manufacturer's guide for appropriate organic solvent

- **Cleaning Materials**
  - ✓ Lint-free cotton gauze pads
  - ✓ Lint-free cotton swabs
  - ✓ Lens paper
- **Alternatives include**:
  - ✓ Fine quality tissue paper
  - ✓ Muslin cloth
  - ✓ Silk

B. procedure to Microscope Cleaning

1. Cleaning the Eyepiece
2. Cleaning the Objectives
3. Cleaning the Microscope Stage
4. Cleaning the Microscope Body
5. Cleaning the Condenser

Step 1: Cleaning the Eyepieces

- ✓ Blow to remove dust before wiping lens
- ✓ Clean the eyepieces with a cotton swab moistened with lens cleaning solution
- ✓ Clean in a circular motion inside out

Step 2: Cleaning the Objectives

- ✓ Objectives are cleaned while attached to microscope
- ✓ Moisten the lens paper with the cleaning solution
- ✓ Wipe gently the objective in circular motion from inside out
- ✓ Wipe with dry tissue or lens cleaning paper
- ✓ Objectives should never be removed from the nosepiece.

Step 3: Cleaning the Microscope Stage

- ✓ Wipe the microscope stage using the cleaning solution on a soft cloth
- ✓ Thoroughly dry the stage
- ✓ Repeat above steps, if required

Step 4: Cleaning the Microscope Body

- ✓ Unplug the microscope from power source
- ✓ Moisten the cotton pad with a mild cleaning agent
- ✓ Wipe the microscope body to remove dust, dirt, and oil
- ✓ Repeat steps1–3, if required

Step 5: Cleaning the Condenser and Auxiliary Lens

- ✓ Unplug the microscope from power source
- ✓ Clean the condenser lens and auxiliary lens using lint-free cotton swabs moistened with lens cleaning solution
- ✓ Wipe with dry swabs

II. Microscope Storage

- ✓ Proper storage of the microscope will prevent or reduce problems!
- ✓ Optics and mechanisms of the microscope must be protected from:
  - •Dust and dirt
  - •Fungus

- **Store the microscope**
  - •Under a protective cover
  - •In a low humidity environment

LAP TEST-2	Performance Test

Name………………………………. ID……………………………..

Date…………………………………….

Time started: ________________________ Time finished: ________________

Instructions: To perform cleaning tools, considerate to check presence of all required facility which is listed in information sheet and operation sheet carefully.

Task-1 Perform how to clean and disinfect basic veterinary tools and equipments

Task 2. Clean all parts of the microscope

LG #3	LO #3- Handle and store basic tools and equipment

Instruction sheet 3.
This learning guide is developed to provide you the necessary information regarding the following content coverage and topics: • Handling of cleaning materials • Safe handling and storing basic veterinary tools and equipment • Reporting work outcomes This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to: • Store or dispose Cleaning materials according to manufacturer's instructions • Maintain and store Tools and equipment according to manufacturers' specifications. • Report the Work outcomes to supervisors
Learning Instructions:
1. Read the specific objectives of this Learning Guide. 2. Follow the instructions described below. 3. Read the information written in the information Sheets 4. Accomplish the Self-checks 5. Perform Operation Sheets 6. Do the "LAP test"

Information Sheet 3:

3.1 Handling of cleaning materials (chemical)

When used properly, both conventional and green cleaning chemicals are relatively safe. However, accidents do happen and there are precautions that should be taken for the proper handling of cleaning solutions.

Here are the key components of a cleaning chemical safety program:

- A complete list of all cleaning chemicals used in the facility; this documentation should include details such as how many gallons (and multiple-gallon containers) are stored, where they are stored and the potential hazards of and necessary precautions for each specific chemical (for instance, whether or not a chemical needs to be kept away from direct sunlight).
- Safety data sheets (SDS) for each chemical used or stored.
- Keeping all cleaning chemicals in their original containers and never mixing chemicals, even if they are the same "type" of chemical.
- Storing chemicals in well-ventilated areas away from HVAC intake vents. This helps prevent any fumes from spreading to other areas of the facility.
- Installing safety signage in multiple languages (or, even better, using images and no words) that quickly conveys possible dangers and precautions related to the chemicals.
- Making sure all cleaning workers know exactly what the following "signal words" mean:
  - ✓ Caution - the product should be used carefully but is relatively safe;
  - ✓ Warning - the product is moderately toxic;
  - ✓ Danger - the product is highly toxic and may cause permanent damage to skin and eyes.

Cleaning chemical safety programs should also include getting rid of chemicals that have not been used for a prolonged period of time," says Jennifer Meek, director of marketing and customer relations for Environmental Solutions. "A good rule of thumb is to consider disposing of any chemical product that has not been used for 6 months, and disposing of any product that has not been used for a year."

3.2 Safe handling and storing basic veterinary tools and equipment

3.2.1. Safe handling of veterinary tools and equipments

The way tools and equipment are handled and stored may be an afterthought to many manufacturers more so when the real driver of a company's success relies on the productivity and efficiency of its processes. However, improper tool handling can impact both those key performance indicators. As most tools, can be fragile and susceptible to damage which inevitably lead to increased machine downtime, quality defects, and dampened profits. To avoid this, the solution could be as simple as proper tool handling

Workers should be trained on safe procedures for working with tools. However, safe practices when carrying or storing those tools may not be thoroughly covered. Tools can pose a safety risk when they are misplaced or improperly handled by workers. The National Safety Council offers the following tips for safe handling of tools when they are not in use:

- Workers should never carry tools up or down a ladder in a way that inhibits grip. Ideally, tools should be hoisted up and down using a bucket or strong bag, rather than being carried by the worker.
- Tools should always be carefully handled from one employee to another never tossed. Pointed tools should be passed either in their carrier or with the handles toward the receiver.
- Workers carrying large tools or equipment on their shoulders should pay close attention to clearances when turning and maneuvering around the workplace.
- Tools should always be put away when not in use. Leaving tools lying around on an elevated structure such as a scaffold poses a significant risk to workers below. This risk increases in areas with heavy vibration.

3.2.2. Veterinary tools and equipment Storage tip

By taking proper care of your tools, you'll ensure that they'll remain in good working order and will be ready for use when you need them. No matter what kind of tools currently in your possession, it's important to take some time organizing your collection so you're protecting your investment. You'll want them in good condition when it's time to start that next DIY project!

If you're running out of room in your home's storage spaces, consider renting a self storage unit from Self Storage Specialists. Putting tools, especially larger tools like power saws, in self storage will keep them safe and away from children. Self storage is commonly used by contractors, too, because it allows them a place to store their equipment while keeping overhead costs low.

Regardless of where you choose to store your tools, there are a few basic tool storage ideas and tips to keep in mind before you put them away.

- **Follow the instructions.** Some manufacturers will have specific instructions for how to store tools, so consult your manual first and foremost. It's important to follow these instructions, especially for larger power tools like saws or drills, so they remain in good working condition.

- **Clean them off.** Tools should be cleaned each time you use them. Wipe them down with a damp rag or towel to get rid of any dirt, dust, grease or debris. Make sure garden tools are free of mud and grime. Everything should be completely dry before placing it in storage to avoid rust developing.

- **Use original cases.** Power tools usually come in hard, plastic cases, and it's recommended to keep these cases for storage whenever possible. These cases will keep your power tools in storage safe from extreme conditions, plus all the parts can be stored right alongside them in the case. No more lost power cords or chargers!

- **Invest in sturdy storage containers.** If you don't have the original container, or you're storing smaller hand tools, invest in some sturdy containers. This will not only keep your tools organized, but also allows them to be easily transportable to your next project area.

- **Store in a safe, dry place.** Along with having the right containers, another way to protect your tools is to ensure that area you're storing them in is safe and dry. Water or humidity can cause damage to tools, especially power tools.

- **Go vertical.** Tools should never be stored on the ground. Invest in some shelving for smaller tools, or hang pegboard along your workbench or on a wall in your garage.

You'll be able to hang things like wrenches, hammers, box cutters, garden equipment and many other tools so they'll be easy to access at any time.

Generally Reprocessing and safe handling of medical device comprises of;

- Preparation(pre-treatment, collecting, pre-cleaning and where applicable, taking the instrument apart)
- Cleaning , disinfecting, final rinse, drying(if required)
- Visual inspection of cleanliness and soundness of material
- Care and repair where required
- Functional testing
- Marking
- Where applicable, packaging and sterilizing, approval for re-use and storage.

3.3 Reporting Work Outcomes

As an independent statutory body, we perform our functions impartially and in an accountable and transparent manner. As part of our commitment to accountability and transparency, we publicly report on the outcomes of our various compliance activities we undertake
animal health care and control activities for farm animals- animal health care and control activities for pet animals These activities are carried out by qualified veterinarians when working in veterinary hospitals as well as when visiting farms, kennels or homes, in own consulting and surgery rooms or elsewhere.

Among a veterinary activities going to be reported for a concerned body are include the following components:

- Report the efficiency of working activity of the organization weather it is good or bad.
- If it is good, include the recommendation for its further progress has to be included in report. When, it reported, all departments of the organization have to have prepare a report on their weakness and strength, to give feedback for the organization
- Availability of tools and equipment required to do any task in any veterinary institution
- Identifying and isolating the functional tools and equipment from non functional one and report to the concerned body

- Report the condition of working environments whether it is safe for work and or not
- Reporting the way of unsafe or non functional tools and equipment material are disposed off or some of them are maintained for reprocessing.

Self-Check – 3	**Written test**

Name………………………………………… ID………………………… Date…….

Directions: Answer all the questions listed below.

Test I: Multiple choice (1 point each)

1. Among the following which one is not true about safe handling of tools

   A, Carrying tools up and down C, Throwing of tools to other employee

   B, Put in tools safely in absence work D, Non Vending

2. Which one is the not the best way of handling cleaning materials

   A, keep away from sun C, store in ventilated area

   B, use safety data sheet D, store in overcrowded area

3. Which one is not conducted during storing of tools and equipments?

   A, cleaning C, Store on wait place

   B, Follow instruction D, find storage container

Test II: Short Answer Questions (3 point each)

1. Write the role of proper handling of cleaning agents
2. List the steps to store tools properly
3. What is the role of report writing

Note: **Satisfactory rating – 6 points** **Unsatisfactory - below 6 points**
You can ask you teacher for the copy of the correct answers.

Operation Sheet -3

3.1.How you handle and store any tools

A. Tools and equipments

- ✓ Material going to be handled (e.g., microscope)
- ✓ Cleaning agents (xyline cotton, savlon, alcohol, etc)
- ✓ Shelve or finding storage container

B. Procedures/Steps/Techniques

Step 1. Preparation (pre-treatment, collecting, pre-cleaning and where applicable, taking the instrument apart)

Step 2. Cleaning, disinfecting, final rinse, drying(if required)

Step 3. Visual inspection of cleanliness and soundness of material

Step 4. Care and repair where required

Step 5. Functional testing

Step 6. Marking

Step 7. Where applicable, packaging and sterilizing, approval for re-use and storage.

LAP TEST-3	Performance Test

Name………………………………. ID……………………………..

Date…………………………………….

Time started: _ ___Time finished: ________________

Instructions: Given necessary templates, tools and materials you are required to perform the following tasks within **1** hour. The project is expected from each student to do it.

Task-1 Store the tools properly

Reference Materials

Books and journals

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The experts who developed the learning guide

No	Name	Qualification	Educational background	Region	Phone number	E-mail
1	Dr. Sileshi Aregahagn	DVM, MSc	Veterinary Medicine	Kombolcha/Amhara	0920480599	kochasile@gmail.com
2	Dr. Chemere Ayenew	DVM, MSc Candidate	Veterinary Medicine	Alage/Federal	0913628734	chemereaz@gmail.com
3	Dr. Addisu Bedashu	DVM, MSc	Veterinary Medicine	Holeta/Oromia	0910281160	addisubedashu@gmail.com
4	Dr. Degu Fitehanegest	DVM, MSc	Veterinary Medicine	Alage/Federal	0910525106	degufe@gmail.com
5	**Dr. Dirshaye Kebede**	**DVM, MSc Candidate**	**Veterinary Medicine**	**Alage/Federal**	**0916497739**	**dirshaye.kebede@yahoo.com**
6	Mr. Amsalu Bedasso	MSc	RDAE	Alage/Federal	0911353949	Amsalub5@gmail.com
7	Mr. Derebe Tesema	BVSc, MSc Candidate	Animal health	Alage/Federal	0982253307	derebetesema08@gmail.com